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(54) **SOLID FORMS OF AN HIV CAPSID INHIBITOR**

FESTE FORMEN EINES HIV-CAPSID-INHIBITORS

FORMES SOLIDES D'UN INHIBITEUR DE CAPSIDE DU VIH

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(73) Proprietor: **Gilead Sciences, Inc.
Foster City, CA 94404 (US)**

(72) Inventor: **SHI, Bing
Foster City
CA 94404 (US)**

(74) Representative: **Carpmaels & Ransford LLP
One Southampton Row
London WC1B 5HA (GB)**

(56) References cited:
**WO-A1-2016/033243 WO-A1-2018/035359
WO-A1-2018/145021 WO-A2-2014/134566**

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Description

TECHNICAL FIELD

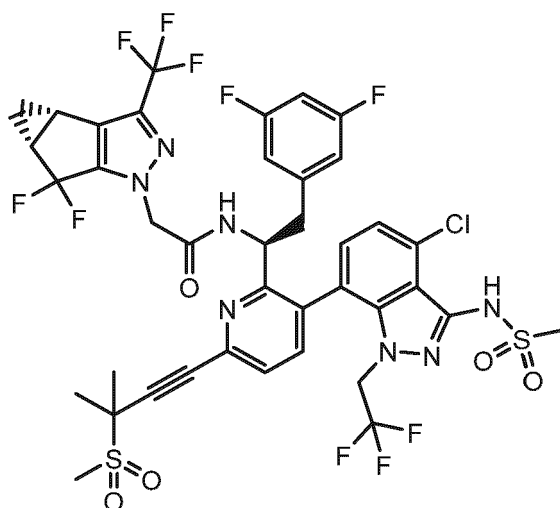
[0001] The present disclosure relates to solid forms of an HIV capsid inhibitor, including pharmaceutically acceptable salts in crystalline form and cocrystals of the inhibitor, for use in the treatment of a *Retroviridae* viral infection including an infection caused by the HIV virus. The present disclosure also relates to pharmaceutical compositions containing the novel salts in crystalline form and cocrystals thereof, and their use in methods of treating or preventing a *Retroviridae* viral infection.

BACKGROUND

[0002] Positive-single stranded RNA viruses comprising the Retroviridae family include those of the subfamily Orthoretrovirinae and genera Alpharetrovirus, Betaretrovirus, Gammaretrovirus, Deltaretrovirus, Epsilonretrovirus, Lentivirus, and Spumavirus which cause many human and animal diseases. Among the Lentivirus, HIV-1 infection in humans leads to depletion of T helper cells and immune dysfunction, producing immunodeficiency and vulnerability to opportunistic infections. Treating HIV-1 infections with highly active antiretroviral therapies (HAART) has proven to be effective at reducing viral load and significantly delaying disease progression (Hammer, S.M., et al.; JAMA 2008, 300: 555-570). However, these treatments could lead to the emergence of HIV strains that are resistant to current therapies (Taiwo, B., International Journal of Infectious Diseases 2009, 13:552-559; Smith, R. J., et al., Science 2010, 327:697-701). Therefore, there is an ongoing need to discover new antiretroviral agents and to develop methods for their preparation and purification as well as prepare improved pharmaceutical formulations of the same. The solid forms of the HIV capsid inhibitor disclosed herein help meet these and other needs. WO2016/033243 and WO2014/134566 disclose indazole compounds and their medical use in the treatment of HIV infection.

SUMMARY

[0003] The present application provides various pharmaceutically acceptable salts in crystalline form or cocrystals of Compound 1 as defined in the appended claims:



Compound 1

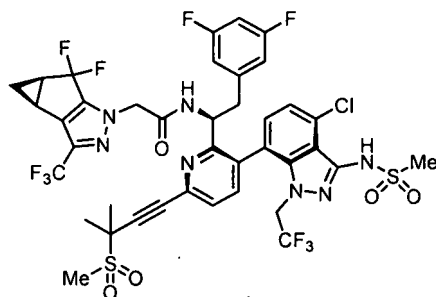
[0004] The present application further provides a pharmaceutical composition comprising the pharmaceutically acceptable salt in crystalline form or cocrystal of Compound 1 disclosed herein, and at least one pharmaceutically acceptable excipient.

[0005] The present application further provides the pharmaceutically acceptable salt in crystalline form or cocrystal of the present invention for use in therapy.

[0006] The present application further provides the pharmaceutically acceptable salt in crystalline form or cocrystal of the present invention for use in a method of treating or preventing a human immunodeficiency virus (HIV) infection, comprising administering a therapeutically effective amount of the salt in crystalline form or the cocrystal to a subject in

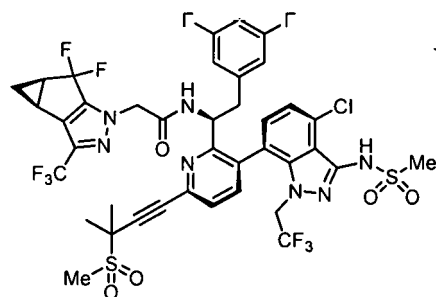
need thereof.

[0007] The present application further provides a process of increasing the amount of an isomeric compound of Isomer A:



Isomer A

relative to an amount of an isomeric compound of Isomer B:



Isomer B

or the amount of an isomeric compound of Isomer B relative to the amount of an isomeric compound of Isomer A, in a starting mixture comprising both isomeric compounds, the process comprising:

contacting the starting mixture with an acid or a base in the presence of a solvent to form a salt mixture of both isomeric compounds, wherein the salt mixture has an increased amount of the isomeric salt of Isomer A relative to the amount of the isomeric salt of Isomer B, or an increased amount of the isomeric salt of Isomer B relative to the amount of the isomeric salt of Isomer A, when compared with the relative amounts of the isomeric compounds of Isomer A and Isomer B in the starting mixture.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008]

Figure 1 shows an XRPD pattern characteristic of the sodium salt of Compound 1, crystalline Form I.

Figure 2 shows a DSC thermogram characteristic of the sodium salt of Compound 1, crystalline Form I.

Figure 3 shows an XRPD pattern characteristic of the sodium salt of Compound 1, crystalline Form II.

Figure 4 shows a DSC thermogram characteristic of the sodium salt of Compound 1, crystalline Form II.

Figure 5 shows an XRPD pattern characteristic of the sodium salt of Compound 1, crystalline Form III.

Figure 6 shows a DSC thermogram characteristic of the sodium salt of Compound 1, crystalline Form III.

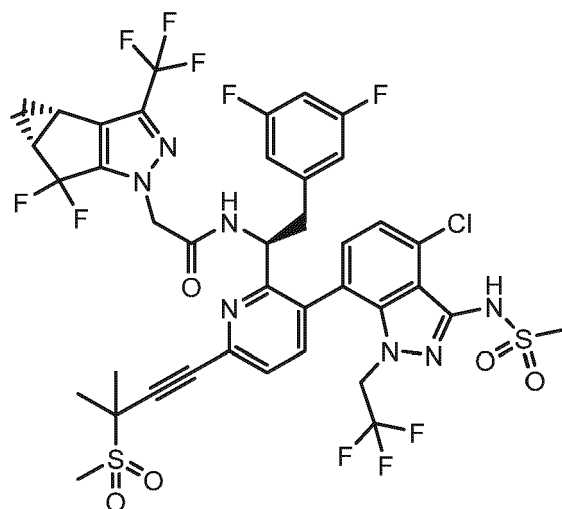
Figure 7 shows an XRPD pattern characteristic of the potassium salt of Compound 1.

Figure 8 shows an XRPD pattern characteristic of the methanesulfonic acid salt or cocrystal of Compound 1, crystalline Form I.

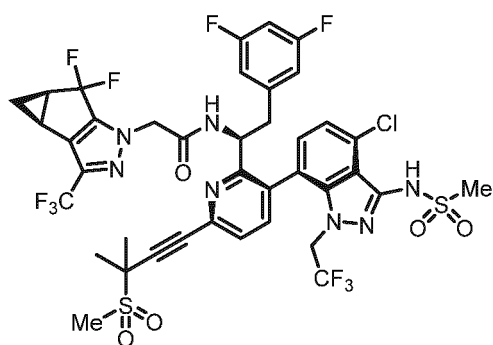
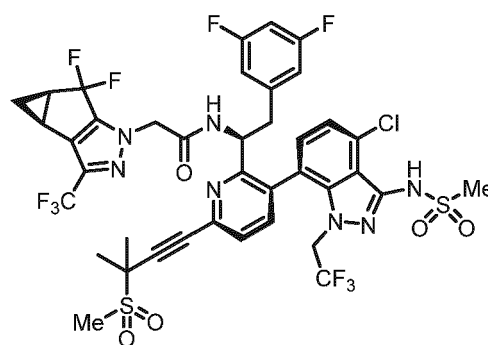
Figure 9 shows a DSC thermogram characteristic of the methanesulfonic acid salt or cocrystal of Compound 1, crystalline Form I.

Figure 10 shows an XRPD pattern characteristic of the methanesulfonic acid salt or cocrystal of Compound 1, crystalline Form II.

Figure 11 shows a DSC thermogram characteristic of the methanesulfonic acid salt or cocrystal of Compound 1, crystalline Form II.

**Compound 1**

[0010] Compound 1 can be a weak acid (pK_a 6.7 at sulfonamide) or a weak base (pK_a <1 at pyridine) and consists of two atropisomers, Isomer A and Isomer B, that can rotate along one of the C-C bonds, as shown below. In solution, the two atropisomers coexist in the ratio of about 1:5 to 1:8 (Isomer A:Isomer B), depending on temperatures and pH. The two atropisomers can be separated by chromatography but they re-equilibrate in solution ($t_{1/2} \approx 1$ -2 hours at 37 °C) and the rotational energy barrier is about 24 kcal/mol.

**Isomer A****Isomer B**

[0011] As described herein, Isomer B can be enriched by crystallization. For example, Isomer B can be enriched by preparing Form I of the sodium salt and Forms I/III of the methanesulfonic acid salt (or cocrystal).

[0012] The solid forms of the invention include crystalline salt forms as well as cocrystal forms of Compound 1. In further embodiments, the solid form can be a cocrystal of Compound 1, in which Compound 1 has formed a crystalline solid together with a coformer molecule. Both crystalline salts and cocrystals of Compound 1 can exist in different crystalline forms (i.e., have different polymorphic or pseudopolymorphic forms).

[0013] As used herein, the term "cocrystal" refers to a compound (such as Compound 1) crystallized together with one or more coformer molecules (e.g., molecules other than the compound). Depending on the chemical nature and proportion of cofomers present in the cocrystal, different physical properties related to, for example, dissolution and solubility may be observed compared with solid forms of the compound by itself or salts thereof. In some instances, the coformer molecule may be a protic acid, and whether the protic acid forms a salt or a cocrystal will often depend on the relative pK_a 's of the compound and coformer. See, e.g., Regulatory Classification of Pharmaceutical CoCrystals: Guidance for Industry, revised August 2016, published by the U.S. Dept. of Health and Human Services, FDA, Center for Drug Evaluation and Research (CDER).

[0014] As used herein, "crystalline form" is meant to refer to a certain lattice configuration of a crystalline substance (e.g., a salt or a cocrystal). Different crystalline forms of the same substance typically have different crystalline lattices (e.g., unit cells) which are attributed to different physical properties that are characteristic of each of the crystalline forms. In some instances, different lattice configurations have different water or solvent content.

[0015] According to the present invention, a crystalline form of a salt or cocrystal of Compound 1 can be useful in the

terized by a DSC thermogram substantially as shown in Figure 24.

[0075] In some embodiments, crystalline Form III of the hydrochloric acid salt or cocrystal of Compound 1 is characterized by a DSC thermogram having a melting onset of about 112 °C.

5 *Compound 1 sulfuric acid salt or cocrystal*

[0076] Compound 1 can be isolated as a sulfuric acid salt or cocrystal which is crystalline.

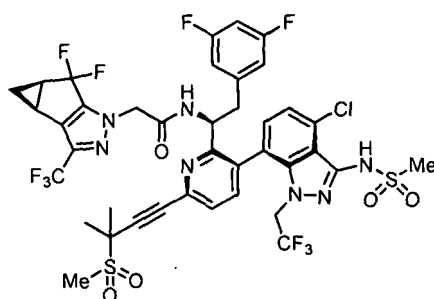
[0077] In some embodiments, the crystalline sulfuric acid salt or cocrystal of Compound 1 has an XRPD profile substantially as shown in Figure 25.

10 **[0078]** In some embodiments, the crystalline sulfuric acid salt or cocrystal of Compound 1 has at least one, at least two, at least three, at least four, at least five, at least six, at least seven, at least eight, or at least nine XRPD peaks, in terms of $2\text{-theta} \pm 0.2^\circ$, selected from 14.2° , 15.3° , 16.3° , 18.3° , 19.1° , 19.3° , 22.6° , 23.9° , and 27.7° .

[0079] In some embodiments, the crystalline sulfuric acid salt or cocrystal of Compound 1 is characterized by a DSC thermogram substantially as shown in Figure 26.

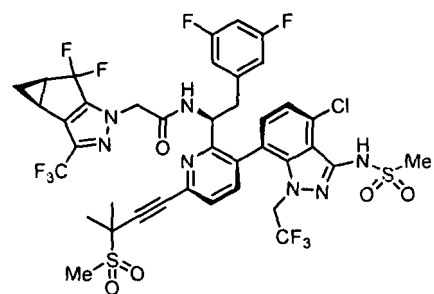
15 **[0080]** In some embodiments, the crystalline sulfuric acid salt or cocrystal of Compound 1 is characterized by a DSC thermogram having a melting onset of about 169 °C.

[0081] The present application further provides a process of increasing the amount of an isomeric compound of Isomer A:



Isomer A

relative to an amount of an isomeric compound of Isomer B:



Isomer B

or the amount of an isomeric compound of Isomer B relative to the amount of an isomeric compound of Isomer A, in a starting mixture comprising both isomeric compounds, the process comprising:

contacting the starting mixture with an acid or a base in the presence of a solvent to form a salt mixture of both isomeric compounds, wherein the salt mixture has an increased amount of the isomeric salt of Isomer A relative to the amount of the isomeric salt of Isomer B, or an increased amount of the isomeric salt of Isomer B relative to the amount of the isomeric salt of Isomer A, when compared with the relative amounts of the isomeric compounds of Isomer A and Isomer B in the starting mixture.

[0082] As used herein, the term "contacting" refers to the bringing together of the indicated moieties, e.g., in a mixture.

[0083] In some embodiments, the process comprises increasing the amount of an isomeric compound of Isomer A relative to an amount of an isomeric compound of Isomer B. In some embodiments, the process of increasing the amount of an isomeric compound of Isomer A relative to an amount of an isomeric compound of Isomer B comprises contacting the starting mixture with a base in the presence of a solvent.

carried forth in the next synthetic step.

Formulation Example A.

[0274] A suspension of Form I of the sodium salt of Compound 1 in 2% poloxamer 188 in saline (200mg/mL) was prepared. The suspension was administered to dogs subcutaneously at a dose of 6mg/kg and the pharmacokinetic (PK) profile was determined. Figure 27 shows a plot of the plasma concentration of Compound 1 as a function of time. As Figure 27 shows, Compound 1 has measurable plasma concentrations at day 70 demonstrating extended release pharmacokinetics.

Formulation Example B.

[0275] A solution of the Form I of the sodium salt of Compound 1 in NMP (200mg/mL) was prepared. The solution was administered to dogs subcutaneously at a dose of 6mg/kg and the pharmacokinetic (PK) profile was determined. Figure 28 shows a plot of the plasma concentration of Compound 1 as a function of time. As the data shows in Figure 28, Compound 1 has measurable plasma concentrations at day 70 demonstrating extended release pharmacokinetics.

Formulation Example C.

[0276] A solution formulation containing 200mg/mL of Compound 1 with 1.2 molar equivalent NaOH to form *in situ* sodium salt in 10% ethanol, 13% water, and 77% PEG are provided. Subjects were orally dosed with this formulation at 6mg/kg. A solution of the Compound 1 in 10% ethanol, 13% water, and 77% PEG 200 (200mg/mL) with 1.2 molar equivalent NaOH was prepared to form *in situ* sodium salt. The solution was administered to dogs subcutaneously at a dose of 6mg/kg and the pharmacokinetic (PK) profile was determined. Figure 29 shows a plot of the plasma concentration of Compound 1 as a function of time. As the data shows in Figure 29, Compound 1 has measurable plasma concentrations at day 28 demonstrating extended release pharmacokinetics.

Formulation Example D.

[0277] A solution formulation of Compound 1 in 10% ethanol, 13% water, and 77% glycofurol (200 mg/mL) with 1.2 molar equivalent NaOH was prepared to form *in situ* sodium salt. The solution was administered to dogs subcutaneously at a dose of 6 mg/kg and the pharmacokinetic (PK) profile was determined. Figure 30 shows a plot of the plasma concentration of Compound 1 as a function of time. As the data shows in Figure 30, Compound 1 has measurable plasma concentrations at day 28 demonstrating extended release pharmacokinetics.

Claims

1. A pharmaceutically acceptable salt, which is *N*-((S)-1-(3-(4-chloro-3-(methylsulfonamido)-1-(2,2,2-trifluoroethyl)-1H-indazol-7-yl)-6-(3-methyl-3-(methylsulfonyl)but-1-yn-1-yl)pyridin-2-yl)-2-(3,5-difluorophenyl)ethyl)-2-((3bS,4aR)-5,5-difluoro-3-(trifluoromethyl)-3b,4,4a,5-tetrahydro-1H-cyclopropa[3,4]cyclopenta[1,2-c]pyrazol-1-yl)acetamide sodium salt, wherein the pharmaceutically acceptable salt is in crystalline form.

2. The pharmaceutically acceptable salt in crystalline form of claim 1, wherein the crystalline Form is

a) crystalline Form I, wherein the crystalline Form I has at least three XRPD peaks, in terms of 2-theta $\pm$ 0.2°, selected from 5.6°, 6.6°, 10.9°, 13.4°, 16.8°, 17.1°, 21.8°, 24.1°, and 26.9° as measured at a radiation wavelength of 1.5406 Å, optionally wherein the crystalline Form I is **characterized by** a DSC thermogram having a melting onset of 218 °C as measured at a heating rate of 10 °C/min; or

b) crystalline Form II, wherein the crystalline Form II has at least three XRPD peaks, in terms of 2-theta $\pm$ 0.2°, selected from 5.4°, 7.0°, 11.1°, 17.7°, 19.2°, 21.2°, 22.6°, 24.0°, and 27.7° as measured at a radiation wavelength of 1.5406 Å, optionally wherein the crystalline Form II is **characterized by** a DSC thermogram having a melting onset of 222 °C as measured at a heating rate of 10 °C/min; or

c) crystalline Form III, wherein the crystalline Form III has at least three XRPD peaks, in terms of 2-theta $\pm$ 0.2°, selected from 5.9°, 7.1°, 11.6°, 15.4°, 17.2°, 18.4°, 19.5°, 22.2°, and 27.2° as measured at a radiation wavelength of 1.5406 Å, optionally wherein the crystalline Form III is **characterized by** a DSC thermogram having a melting onset of 213 °C as measured at a heating rate of 10 °C/min.

3. A pharmaceutically acceptable salt or cocrystal, which is a *N*((*S*)-1-(3-(4-chloro-3-(methylsulfonamido)-1-(2,2,2-trifluoroethyl)-1H-indazol-7-yl)-6-(3-methyl-3-(methylsulfonyl)but-1-yn-1-yl)pyridin-2-yl)-2-(3,5-difluorophenyl)ethyl)-2-((3*bS*,4*aR*)-5,5-difluoro-3-(trifluoromethyl)-3*b*,4,4*a*,5-tetrahydro-1H-cyclopropa[3,4]cyclopenta[1,2-*c*]pyrazol-1-yl)acetamide methanesulfonic acid salt or cocrystal, wherein the pharmaceutically acceptable salt is in crystalline form.

4. The salt of claim 3 in crystalline form, wherein the crystalline form is hydrated or solvated.

5. The crystalline form of claim 3 or claim 4, wherein the crystalline Form is

a) crystalline Form I, wherein the crystalline Form I has at least three XRPD peaks, in terms of 2-theta $\pm$ 0.2°, selected from 12.9°, 15.4°, 18.4°, 18.8°, 19.7°, 20.2°, 20.9°, 23.5°, and 25.3° as measured at a radiation wavelength of 1.5406 Å, optionally wherein the crystalline Form I is **characterized by** a DSC thermogram having a melting onset of 130 °C as measured at a heating rate of 10 °C/min; or

b) crystalline Form II, wherein the crystalline Form II has at least three XRPD peaks, in terms of 2-theta $\pm$ 0.2°, selected from 8.7°, 13.0°, 17.5°, 19.3°, 20.6°, 21.3°, 21.7°, 24.2°, and 25.3° as measured at a radiation wavelength of 1.5406 Å, optionally wherein the crystalline Form II is **characterized by** a DSC thermogram having a melting onset of 165 °C as measured at a heating rate of 10 °C/min; or

c) crystalline Form III, wherein the crystalline Form III has at least three XRPD peaks, in terms of 2-theta $\pm$ 0.2°, selected from 8.2°, 11.3°, 12.8°, 15.7°, 16.9°, 20.1°, 21.8°, 22.6°, and 24.7° as measured at a radiation wavelength of 1.5406 Å, optionally wherein the crystalline Form III is **characterized by** a DSC thermogram having a melting onset of 141 °C as measured at a heating rate of 10 °C/min; or

d) crystalline Form IV hydrate, wherein the crystalline Form IV hydrate has at least three XRPD peaks, in terms of 2-theta $\pm$ 0.2°, selected from 7.9°, 11.1°, 12.1°, 12.7°, 16.9°, 21.2°, 21.7°, 25.4°, and 26.6° as measured at a radiation wavelength of 1.5406 Å.

6. A pharmaceutically acceptable salt or cocrystal, which is a *N*((*S*)-1-(3-(4-chloro-3-(methylsulfonamido)-1-(2,2,2-trifluoroethyl)-1H-indazol-7-yl)-6-(3-methyl-3-(methylsulfonyl)but-1-yn-1-yl)pyridin-2-yl)-2-(3,5-difluorophenyl)ethyl)-2-((3*bS*,4*aR*)-5,5-difluoro-3-(trifluoromethyl)-3*b*,4,4*a*,5-tetrahydro-1H-cyclopropa[3,4]cyclopenta[1,2-*c*]pyrazol-1-yl)acetamide ethanesulfonic acid salt or cocrystal, wherein the pharmaceutically acceptable salt is in crystalline form.

7. The salt of claim 6 in crystalline form, wherein said crystalline form has at least three XRPD peaks, in terms of 2-theta $\pm$ 0.2°, selected from 9.1°, 12.7°, 13.3°, 15.2°, 18.8°, 19.5°, 20.5°, 22.4°, and 25.3° as measured at a radiation wavelength of 1.5406 Å, optionally wherein the crystalline form is **characterized by** a DSC thermogram having a melting onset of 119 °C as measured at a heating rate of 10 °C/min.

8. A pharmaceutically acceptable salt or cocrystal, which is a *N*((*S*)-1-(3-(4-chloro-3-(methylsulfonamido)-1-(2,2,2-trifluoroethyl)-1H-indazol-7-yl)-6-(3-methyl-3-(methylsulfonyl)but-1-yn-1-yl)pyridin-2-yl)-2-(3,5-difluorophenyl)ethyl)-2-((3*bS*,4*aR*)-5,5-difluoro-3-(trifluoromethyl)-3*b*,4,4*a*,5-tetrahydro-1H-cyclopropa[3,4]cyclopenta[1,2-*c*]pyrazol-1-yl)acetamide benzenesulfonic acid salt or cocrystal, wherein the pharmaceutically acceptable salt is in crystalline form.

9. The salt of claim 8 in crystalline form, wherein said crystalline form has at least three XRPD peaks, in terms of 2-theta $\pm$ 0.2°, selected from 7.5°, 8.5°, 13.6°, 17.0°, 18.5°, 18.9°, 20.0°, 21.7°, and 26.6° as measured at a radiation wavelength of 1.5406 Å.

10. A pharmaceutically acceptable salt or cocrystal, which is a *N*((*S*)-1-(3-(4-chloro-3-(methylsulfonamido)-1-(2,2,2-trifluoroethyl)-1H-indazol-7-yl)-6-(3-methyl-3-(methylsulfonyl)but-1-yn-1-yl)pyridin-2-yl)-2-(3,5-difluorophenyl)ethyl)-2-((3*bS*,4*aR*)-5,5-difluoro-3-(trifluoromethyl)-3*b*,4,4*a*,5-tetrahydro-1H-cyclopropa[3,4]cyclopenta[1,2-*c*]pyrazol-1-yl)acetamide hydrochloric acid salt or cocrystal, wherein the pharmaceutically acceptable salt is in crystalline form.

11. The salt of claim 10 in crystalline form, wherein the crystalline Form is

a) crystalline Form I, wherein the crystalline Form I has at least three XRPD peaks, in terms of 2-theta $\pm$ 0.2°, selected from 9.4°, 12.6°, 14.3°, 15.4°, 16.4°, 20.1°, 21.6°, 22.5°, and 23.8° as measured at a radiation wavelength of 1.5406 Å, optionally wherein the crystalline Form I is **characterized by** a DSC thermogram having a

melting onset of 101 °C as measured at a heating rate of 10 °C/min; or

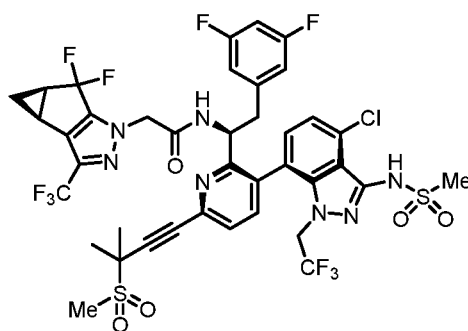
b) crystalline Form II, wherein the crystalline Form II has at least three XRPD peaks, in terms of 2-theta $\pm$ 0.2°, selected from 17.2°, 17.6°, 18.1°, 18.8°, 22.3°, 22.6°, 23.1°, 25.5°, and 26.9° as measured at a radiation wavelength of 1.5406 Å; or

c) crystalline Form III, wherein the crystalline Form III has at least three XRPD peaks, in terms of 2-theta $\pm$ 0.2°, selected from 9.7°, 12.9°, 16.7°, 17.6°, 18.0°, 18.5°, 19.3°, 22.1°, and 25.0° as measured at a radiation wavelength of 1.5406 Å, optionally wherein the crystalline Form III is **characterized by** a DSC thermogram having a melting onset of 112 °C as measured at a heating rate of 10 °C/min.

12. A pharmaceutically acceptable salt or cocrystal, which is a *N*((S)-1-(3-(4-chloro-3-(methylsulfonamido)-1-(2,2,2-trifluoroethyl)-1H-indazol-7-yl)-6-(3-methyl-3-(methylsulfonyl)but-1-yn-1-yl)pyridin-2-yl)-2-(3,5-difluorophenyl)ethyl)-2-((3bS,4aR)-5,5-difluoro-3-(trifluoromethyl)-3b,4,4a,5-tetrahydro-1H-cyclopropa[3,4]cyclopenta[1,2-c]pyrazol-1-yl)acetamide sulfuric acid salt or cocrystal, wherein the pharmaceutically acceptable salt is in crystalline form.

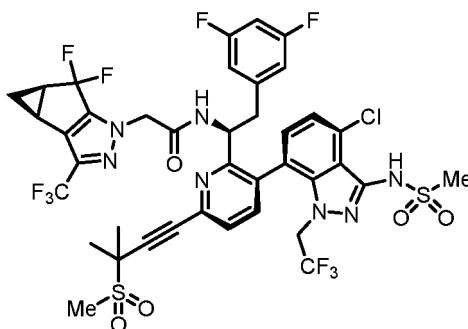
13. The salt of claim 12 in crystalline form, wherein said crystalline form has at least three XRPD peaks, in terms of 2-theta $\pm$ 0.2°, selected from 14.2°, 15.3°, 16.3°, 18.3°, 19.1°, 19.3°, 22.6°, 23.9°, and 27.7° as measured at a radiation wavelength of 1.5406 Å, optionally wherein the crystalline form is **characterized by** a DSC thermogram having a melting onset of 169 °C as measured at a heating rate of 10 °C/min.

14. A preparation comprising the pharmaceutically acceptable salt in crystalline form or the cocrystal of any one of claims 1-13, which is enriched in Isomer A:



Isomer A.

or enriched in Isomer B:



Isomer B.

15. The preparation of claim 14, in which the molar ratio of Isomer A:

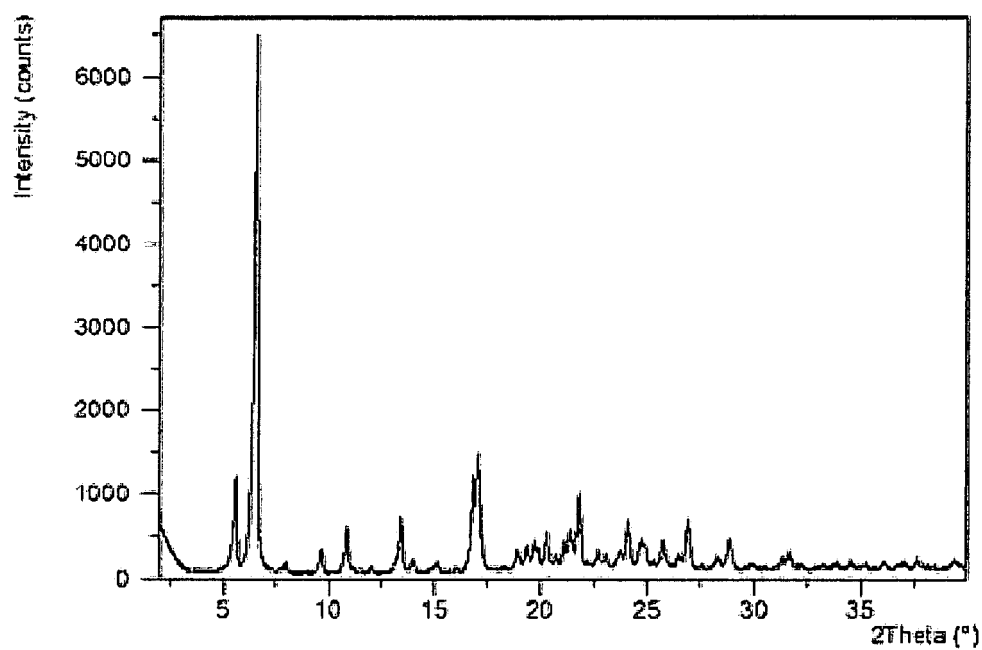


Figure 1

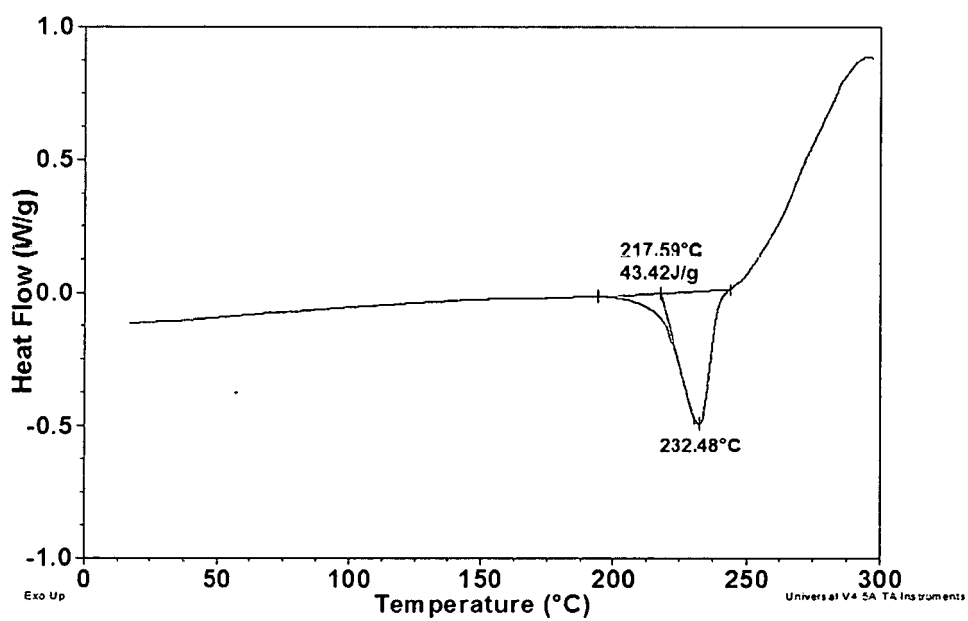


Figure 2